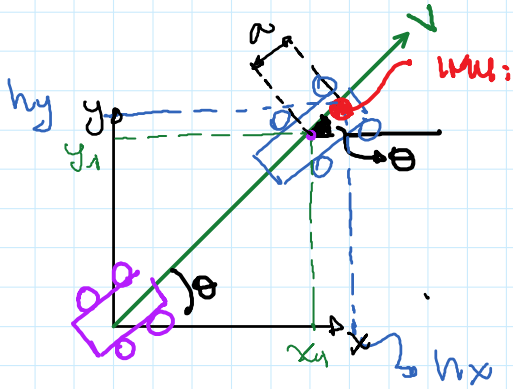
b

martes, 13 de octubre de 2026 19:20

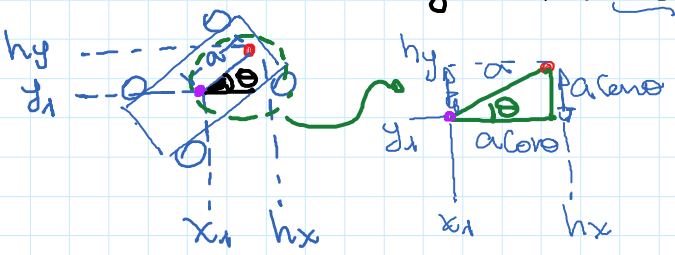
Introducción a la Robótica Móvil (Tercera)

$$\dot{x} = \lim_{\Delta t \to 0} \frac{\Delta x}{\Delta t} = \frac{x(k+1) - x(k)}{\Delta t}$$



UMI: Unidad Inercial de Control.

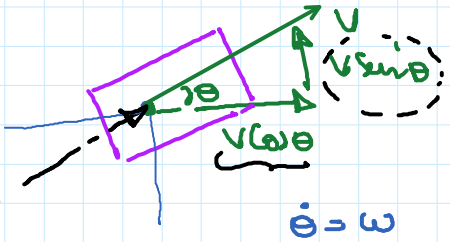
$$h_x = x_1 + a\cos\theta$$
$$h_y = y_1 + a\sin\theta$$



$$\frac{d}{dt}(h_x = x_1 + a\cos\theta)$$
$$\frac{d}{dt}(h_y = y_1 + a\sin\theta)$$

$$\frac{d f(x(\theta))}{dt} = \frac{\partial f(x(\theta))}{\partial \theta} \cdot \frac{\partial \theta}{dt}$$
$$a\cos\theta = a\frac{\partial \cos\theta}{\theta} \cdot \frac{\partial \theta}{\partial t} = -a\sin\theta \cdot \dot{\theta}$$

$$\dot{h}_x = \dot{x}_1 - a\sin\theta\,\dot{\theta}$$
$$\dot{h}_y = \dot{y}_1 + a\cos\theta\,\dot{\theta}$$



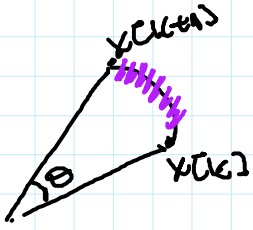
$$\dot{h}_x = V\cos\theta - a\sin\theta\,\dot{\theta}$$
$$\dot{h}_y = V\sin\theta + a\cos\theta\,\dot{\theta}$$

$$\dot{\theta} = \omega$$

Nota: $\Rightarrow \phi = \theta$

$$\begin{bmatrix} \dot{h}_x \\ \dot{h}_y \end{bmatrix} = \begin{bmatrix} \cos\theta & -a\sin\theta \\ \sin\theta & a\cos\theta \end{bmatrix} \begin{bmatrix} V \\ \omega \end{bmatrix}$$

$\dot{p} = J\,\dot{q}$ → Jacobiano.



$$x(k+1) = x(k) + \text{[illegible]}$$

$$\omega = \frac{\theta}{t}$$
$$\theta = \omega t \quad \Big\} \; t = t_s$$

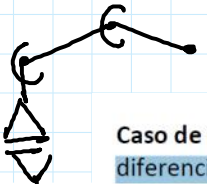


$$x(k+1) = x(k) + \text{[illegible]}$$
$$y(k+1) = y(k) + \text{[illegible]}$$

$$e_x = vt = x_p \cdot t_s$$
$$e_y = y_p \cdot t_s$$

Trabajo Final

Robot Antropomórfico 3gdl

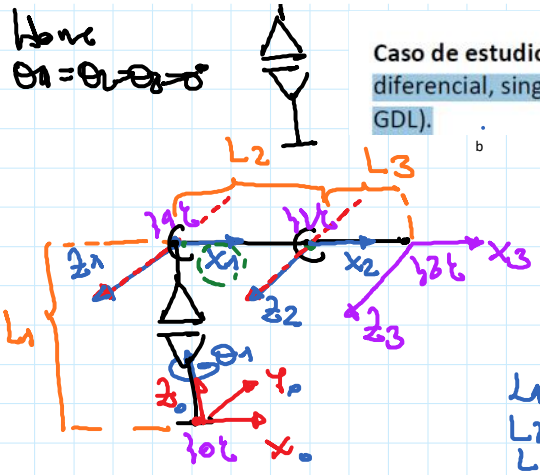


$$\begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

Hoy: $\theta_1 = \theta_2 = \theta_3 = 0°$

Caso de estudio: Desarrollar la simulación de la cinemática directa, cinemática inversa, cinemática diferencial, singularidades, control cinemático y dinámica del robot elegido (antropomórfico de 3

Caso de estudio: Desarrollar la simulación de la cinemática directa, cinemática inversa, cinemática diferencial, singularidades, control cinemático y dinámica del robot elegido (antropomórfico de 3 GDL).



	θ	d	a	α
0-1	θ_1	L_1	0	90°
1-2	θ_2	0	L_2	0°
2-3	θ_3	0	L_3	0°

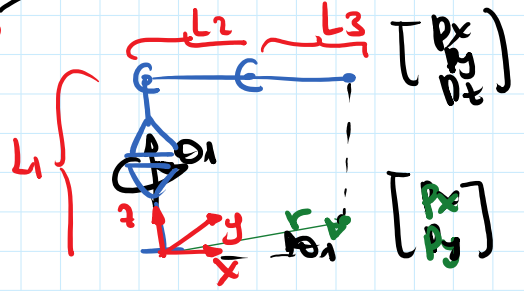
$L_1 = 30\,mm$
$L_2 = 15\,mm$
$L_3 = 10\,mm$

Rango:
$-140° \leq \theta_1 \leq 140°$
$-120° \leq \theta_2 \leq 120°$
$-110° \leq \theta_3 \leq 110°$

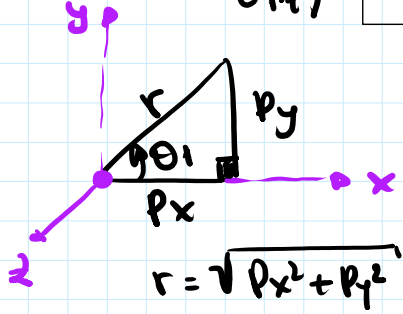
Cinemática Inversa

3D → 2D.

$\begin{bmatrix} p_x \\ p_y \\ p_z \end{bmatrix}$ → Cinemática Inversa → $\begin{bmatrix} \theta_1 \\ \theta_2 \\ \theta_3 \end{bmatrix}$

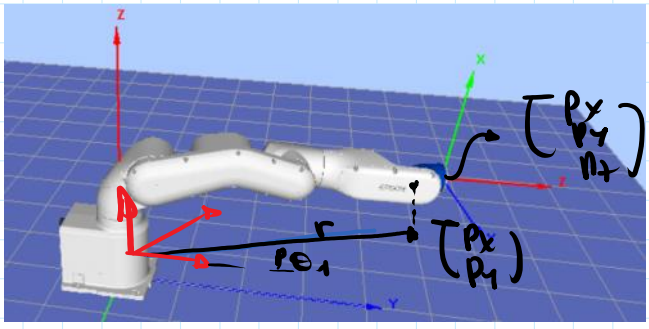


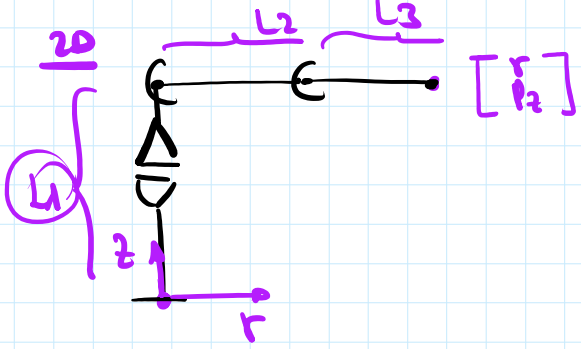
$\begin{bmatrix} p_x \\ p_y \\ p_z \end{bmatrix}$ $\begin{bmatrix} p_x \\ p_y \end{bmatrix}$



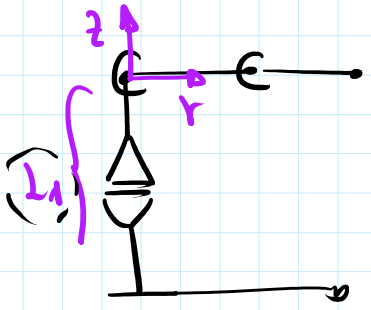
$\theta_1 = \tan^{-1}\left(\frac{p_y}{p_x}\right)$.

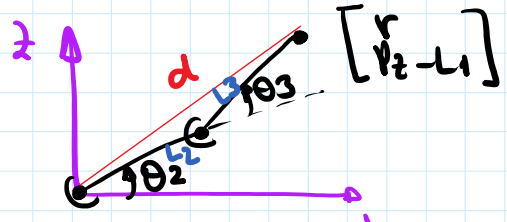
$r = \sqrt{p_x^2 + p_y^2}$ (Pitágoras).





$\begin{bmatrix} r \\ p_z \end{bmatrix}$





$\begin{bmatrix} r \\ p_z - L_1 \end{bmatrix}$

$d = \sqrt{r^2 + (p_z - L_1)^2}$

θ_3:

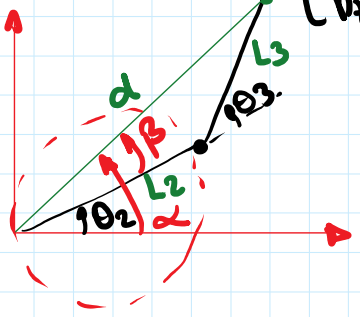
$d^2 = L_2^2 + L_3^2 + 2L_2L_3C_3$

$C_3 = \frac{L_2^2 + L_3^2 - d^2}{2L_2L_3}$

$S_3 = \sqrt{1 - C_3^2}$

$\theta_3 = \tan^{-1}\left(\frac{S_3}{C_3}\right)$

θ_2:



$\begin{bmatrix} r \\ p_z - L_1 \end{bmatrix}$

$\alpha = \tan^{-1}\left(\frac{p_z - L_1}{r}\right)$

$L_3^2 = d^2 + L_2^2 - 2dL_2C_\beta$

$C_\beta = \frac{d^2 + L_2^2 + L_3^2}{2dL_2}$

$S_\beta = \sqrt{1 - C_\beta^2}$

$\beta = \tan^{-1}\left(\frac{S_\beta}{C_\beta}\right)$

$\theta_2 = \alpha - \beta$

¿¿¿

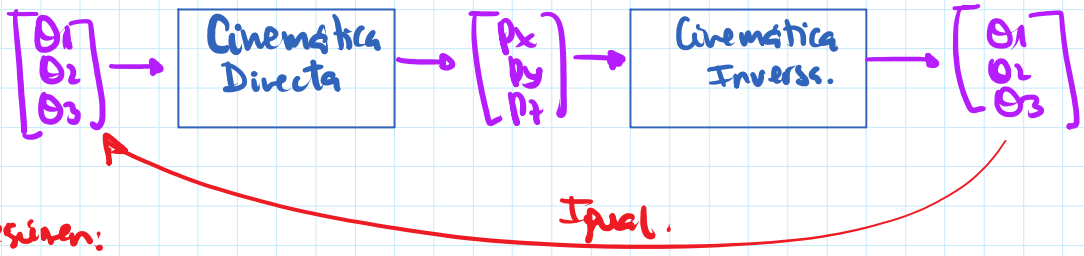
$$S\beta = \sqrt{1-C\beta^2}$$

$$2aL_2$$

$$\theta_2 = \alpha - \beta$$

¿¿¿

Comprobar



Resumen:

1) Cinemática Directa.
2) Espacio de trabajo.
3) Jacobiano lineal.
4) Jacobiano Angular.
5) Singularidad.
6) Cinemática Inversa.

} Robot Antropomórfico de 3GDL.